Satyavada Sri Harini

me23btech11054@iith.ac.in | satyavadaharini@gmail.com | +91 7416447993 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.85
Sri Chaitanya Junior Kalasala	Class XII (Telangana State Board of Intermediate Education)	2023	98.40%
Sri Chaitanya H S, Barkatpura Hyderabad	Class X (Board of Secondary Education, Telangana State, India)	2021	10.00%

PROJECTS

2D Unsteady Heat Conduction Solver (Python, NumPy, C, Numerical Methods)

- Ported a C finite-difference PDE solver to vectorised Python and implemented implicit time integration with three independent linear solvers (ADI with Thomas algorithm, Gauss-Seidel/SOR, Jacobi), cross-validating all three to identical solutions as a correctness control.
- Validated numerical accuracy against an analytically derived conservation invariant, confirming second-order convergence across a three-grid study (64×32 -> 256×128) to within 7×10^(-6).
- Optimised the tridiagonal solve by vectorising the recursion across independent grid lines, and ran parameter studies producing many solution snapshots with automated post-processing and reporting.

Bio-Mimetic Soft Robotic Jellyfish Propulsion

(Under Dr. Prabhat Kumar, Assisstant Professor at IITH)

- Evaluated Physics-Informed Neural Networks (PINNs) on a 1D bar model to capture structural deformation, identifying key limitations in handling dynamically evolving boundary conditions
- Pivoted to a full computational fluid dynamics (CFD)-based fluid-structure interaction (FSI) framework to accurately model the complex locomotive dynamics of the soft robotic structure
- Currently implementing the Immersed Boundary Adaptive Mesh Refinement (IBAMR) C++ framework to couple hyperelastic material behavior with 3D fluid dynamics

Soft Robotics

- *Topology Optimization for Soft Robotic Rehabilitation Glove*
- Optimized the internal pneumatic chamber design of a soft robotic rehabilitation finger using a MATLAB-based topology optimization framework (SoRoTop), formulating a custom surface-tracking objective function to minimize L2 norm error between actuated structural nodes and a target curling trajectory
- Simulated design-dependent fluidic pressure loads via Darcy flow equations and applied MMA (Method of Moving Asymptotes) algorithms to converge on an optimal solid-void distribution, enforcing physical constraints of 45% volume fraction and 15% strain energy to balance actuation flexibility with structural stiffness
- *Bistable Morphing Composite Shells as Grippers*
- Simulated structural bifurcation of a 5-ply composite shell in **Abaqus**, flattening a semi-cylindrical model to initialize internal pre-stress for bistable morphing behavior
- Developed a **Python** script to extract through-thickness bending gradients and stress fields from the Abaqus simulation, feeding them into a flat mesh to initialize accurate stress conditions for the subsequent dynamic snap-through simulation
- Fabricated a TPU physical prototype to replicate composite behavior, validating the rapid transition between flat and stable coiled states

Tohuku Engineering Summer Program (TESP), Sendai, Japan:

- Selected as one of 50 participants in the Robotics track at Tohoku University, ranked first in the Times Higher Education (THE) Japan Rankings.
- Worked in a five-member team to design a reinforcement learning-based solution, training a simulated snake robot to navigate a corridor using the Soft Actor-Critic (SAC) algorithm with Stable-Baselines3 in the MuJoCo simulator.
- Participated in a cultural immersion program, developing a deeper understanding of Japanese traditions and society.

Inter-IIT Tech Meet 14.0 *(Ghost Member, Drona Aviation)*

- Engineered PID control logic and tuning for a flashable, pre-configured drone platform, contributing to a national-level inter-institute technical competition

Inter-IIT Tech Meet 13.0 *(Contingent Member, Albatross Energetics)*

- Built a Simulink/Simscape-based simulation model of an air conditioning system to evaluate and optimize system performance and energy efficiency

Automated Boom Barrier *(BUILD Project)*

- Developed a real-time Python ALPR pipeline for automated access control using YOLO and OpenCV-based OCR preprocessing.
- CAD-modeled the full system protoype, designing custom, precision-aligned camera and Raspberry Pi mounts.
- Fabricated the mechanical assembly using 3D-printed components, seamlessly integrating hardware and software for field deployment.

SKILLS

Programming Languages: C++, Python, Java, SQL

Software & Tools: SolidWorks, SolidEdge, MATLAB, Simulink, Simscape, MSC ADAMS, Arduino IDE

Developer Tools & Practices: Git/GitHub, Linux/Bash

Hardware: ESP32, STM32 microcontrollers, Raspberry Pi 5

Core Competencies: Data Structures & Algorithms (DSA), Machine Learning, Physics-Informed Neural Networks (PINNs), Finite Element Method (FEM), Topology Optimization, Multi-body Dynamics Simulation, Robotics & Control Systems

Frameworks & Libraries: PyTorch, TensorFlow, DeepXDE, MuJoCo

RELEVANT COURSES

Mechanical Core:

Finite Element Method (FEM), Computational Fluid Dynamics (CFD), Modelling and Simulation, Solid Mechanics, Design of Machine Elements, Thermodynamics, Fluid Mechanics, Heat and Mass Transfer, Machine Drawing and Solid Modelling, Composite Structures, Manufacturing Science (1 & 2), IC Engines, Fundamentals of Robotics, Machine Learning and its Applications, Marine Robotics, Soft Robotics

Mathematics:

Calculus - 1, Calculus - 2, Linear Algebra, Differential Equations, Transform Techniques, Probability, Statistics, Complex Variables

Material Science:

Fundamentals of Physical Metallurgy, Mechanical Behaviour of Materials, Composite Materials

POSITIONS OF RESPONSIBILITY

Sponsorship Head, *Aeolus Racing (IITH Racing)*

- Spearheaded the foundational sponsorship strategy for a first-year racing team, successfully securing 3 corporate partnerships to fund vehicle engineering and operations.
- Pitched the team's vision and long-term viability directly to the Institute Director, securing critical institutional funding to expand operations into the team's second year.

Senior Core, *Tinkerers' Lab*

- Mentored a team of 6 junior members in the 3D modeling and conceptualization of a WALL-E robotics prototype using Autodesk Fusion 360 and core engineering design principles.

- Architected the preliminary software framework for the robot, drafting a comprehensive ROS2 integration strategy for future development and deployment.